SEYHMUS GULER

617-504-7329 | seyhmusguler@gmail.com | seyhmusguler.com

SUMMARY

Senior Data Scientist with 10+ years of experience analyzing large-scale datasets and extracting actionable insights. Expert in the Python ecosystem, Bash, Google Cloud Platform, and modern AI tools. Grounded in strong mathematical rigor, I excel at translating highly ambiguous concepts into clearly defined, solvable problems. As a reliable partner across disciplines, I value critical thinking, precise execution, and clear ownership. I have a proven track record of modernizing data infrastructure, shipping innovative data products, and seamlessly retiring legacy systems.

EXPERIENCE

Senior data scientist March 2022 – Present
Recursion Pharmaceuticals *Salt Lake City, UT*
- Builds computational tools to create maps of biology from multi-omics data. Analyzes patient and biological datasets to identify new biological targets for drug discovery. Co-owns and manages several data processing pipelines and products.

Research fellow, *Department of Psychiatry* March 2020 – Feb 2022
Massachusetts General Hospital and Harvard Medical School *Boston, MA*
- Led a project studying the effects of coupled non-invasive brain stimulation and acupuncture on chronic low-back pain. Collected, cleaned, and analyzed functional MRI (fMRI) data using in-house pipelines for various research projects in the lab.

Research fellow, *Department of Radiology* Nov. 2016 – Feb. 2020
Boston Children's Hospital and Harvard Medical School *Boston, MA*
- Developed motion correction and missing-data completion strategies for fMRI and tensor data. Designed and ran an fMRI-neurofeedback protocol to train motor function via neuro-feedback.

Lecturer (part-time), *Department of Electrical and Computer Engineering* June 2016 – Aug. 2019
Northeastern University *Boston, MA*
- Developed the curriculum and taught "Noise and Stochastic Processes" three semesters, and "Introduction to Linear Algebra and Probability for Data Science" one semester. Teaching evaluations available upon request.

Research assistant, *Department of Electrical and Computer Engineering* Sep. 2011 – Oct. 2016
Northeastern University *Boston, MA*
- Formulated and solved various optimization problems for effective current delivery in transcranial direct current stimulation (tDCS) and electrocorticography (ECoG) stimulation.

SOFTWARE SKILLS

Programming languages: Python, Bash, SQL, R
Python libraries: pandas, polars, numpy, scipy, sklearn, pytorch, pandera, pydantic, plotly, seaborn, matplotlib, requests, django, pytest
Tools & Workflows: CI/CD pipelines, testing suites, Google Cloud (gcloud), Git, GitHub, virtual environments, Docker containers, infrastructure as code, LaTeX, Microsoft Office, Mendeley
Neuroimaging Analysis: FSL, Nipype, Freesurfer, fMRIPrep, SPM, CONN, BIDS, ANTs, Nilearn, GIFT
Hands-on experience : Linux workstations, HPC (High-Performance Computing), operating an MRI scanner, tDCS setup and application, EEG data acquisition and analysis

EDUCATION

Ph.D. in Electrical Engineering Sep 2011 – Oct 2016
Northeastern University *Boston, MA*
Thesis title: *Brain Stimulus Pattern Optimization Using Scalp and Cortical Electrode Arrays.* Advisor: *Dana H. Brooks.*

B.Sc. in Electrical and Electronics Engineering Sep 2006 – Jun 2011
Bilkent University *Ankara, Turkey*

AWARDS AND HONORS

Graduate research assistantship (GRA), Northeastern University, Boston, MA (2011-2016)
Full scholarship for undergraduate studies, Bilkent University, Ankara, Turkey (2006-2011)
Ranked 210th in 2006 National University Entrance Exam, Turkey ($\sim$ 1.5 million students)
Ranked 5th (Silver medalist) in 13th National Mathematics Olympiad, TUBITAK, Ankara, Turkey ($\sim$10k students)

PUBLICATIONS

🎓– 7 journal papers (3 first-authored, h-index=7), 3 conference papers, and 6 conference abstracts, all peer-reviewed.

LANGUAGES AND INTERESTS

Languages: Kurdish (Native), Turkish (Native), English (Fluent), Spanish (Beginner)
Interests: Dancing, Racquetball, Snowboarding, Jump rope, Climbing, Hiking, Soccer, Chess, Canyoneering, Mountain biking